1) Roles Overview

Super admin, Employee, HR , Optional (IT / Asset Custodian), Team Leader, Unit Leader, Department Manager, Vice President (optional), CEO

2) Core Data

2.1 Employee Profile

- employee_id
- company_id
- full_name
- phone_number
- address
- email
- department/unit/team
- manager_id
- employment_status
- contract_end_date (if exists)
- insurance_policy_id
- medical_class (A/B/C)
- Salary_package_id
- Company_locations (optional)

2.2 Attendance

- Attendance_id / employee_id
- date
- clock_in_time
- clock_out_time
- status (present/late/absent…)
- location/device_id (optional)
- note

2.3 Notifications / announcement (global)

- Notification_id / recipient_user_id
- type
- title/message
- created_at/read_at

2.4 Survey

- Survey_id / submitter_employee_id
- responses[] (question_id/answer)

- routing_chain[] (role/approver_id/status/comment) /Status

2.5 Complaints

- complaint_id
- employee_id
- category
- description
- attachments[] / status / assignee_hr_id

2.6 Requests

- request_id
- employee_id
- approval_chain[] { "step_order": 1, "role": "TeamLeader", "approver_id": "EMP-2001", "status": "approved", "comment": "OK" },
- status
- reason
- attachments[]
- type
    - Salary slip
    - Hr letter
    - End of Service
    - Resignation
    - Others
        - details
    - Leaves or permissions:
        - leave_type (annual (check balance)/emergency/unpaid/permission/other)
        - start_datetime, end_datetime
        - type
        - duration_hours/days
    - Purchase Order:-
        - items[] (name/qty/unit_price/justification)
        - total_estimate
    - Assets:-
        - asset_type / asset_id
        - request_action (issue/return/replace/damage)
    - Loan:-
        - amount
        - tenor_months
        - repayment_schedule[]
            - installment_id
            - loan_request_id
            - installment_no
            - due_period

        - due_date
        - principal_amount
        - interest_amount
        - total_due
        - updated_date
        - Status:
            - scheduled
            - deducted
            - skipped
            - waived
- Assignment (انتداب):-
    - destination
    - details
    - start_date/end_date (optional)
    - purpose فتره ولا دائم
    - per_diem (اختياري-بدل يومي)

2.7 Salary Package

- salary_package_id
- employee_id
- effective_from (date)
- effective_to (date, optional)
- currency
- basic_salary
- allowances[] (title/amount/type/effective_from/effective_to optional)
- deductions[] (title/amount/type/effective_from/effective_to optional)
- tax[] (title/amount/type/effective_from/effective_to optional)
- insurance_policy_id (for social insurance calculation)
- medical_class (A/B/C) (optional if already on profile)
- payroll_calculation_basis (30_days / working_days)
- status (active/inactive)
- created_at / updated_at
- created_by_hr_id (optional)

2.8 Company

- company_id
- company_name
- Company_logo
- locations
- Yearly_vacation_days
- Company hierarchy

[list of positions and sort them to make sure when send request to be accepted from all with ordering correctly]

3) Application Features By roles:

3.1 SuperAdmin:

- **Manage companies** — create a new company and upload a logo.
- **Configure company hierarchy** — set approval chain order.
- **Manage roles & permissions** — create new users and give them roles.
- **Global announcements** — send a company-wide announcement about a new policy.
- **Monitoring** — review logs for all requests and all companies with its employees and performance.

3.2 Employee

- **View profile** — check phone/address/manager data.
- **Request profile update (phone/address)** — submit new address to HR for approval.
- **Clock in / clock out** — record attendance start/end of day with note (optional).
- **View attendance history** — review month's attendance.
- **Complaints** — submit a complaint with attachments to HR.
- **Complaints -** Track complaint status and assigned HR.
- **Upload requested attachments** — upload pdfs requested by HR.
- **Employee can submit Surveys** and then **Survey follows routing chain** (Team Leader → Unit Leader → Dept Manager → VP optional → CEO, etc. based on hierarchy). And Receive notifications on survey approvals/rejections/comments.
- **View survey history and current status (draft/submitted/in-progress/completed/rejected).**
- **Create requests with many types (Leave/Permission, Salary Slip, HR Letter, Resignation, End of Service, Purchase Order, Assets, Loan, Assignment/انتداب, Others). Then Upload attachments for any request type when required. And Track request status + approval progress (who approved / pending / rejected).**
- **Create leave/permission request** — (check paid or unpaid) request 5 days annual leave; system uses paid balance then unpaid.
- **View all leave types balance & history** — see remaining annual days and past requests.
- **View payslips** — download payslip PDF for a bank request.
- **Receive notifications** — get notified when leave is approved or any notification.
- **Assets request (issue/return/replace/damage)** — request a laptop or report damage with photos.
- **Loan request** — request a 10,000 loan over 6 months.
- **Assignment (انتداب) request** — request a temporary assignment to another site for **custom days**, including an optional **per diem (daily allowance)**.
- **Purchase Order request** — request **Purchase Order** by submitting an **itemized list** (items, quantities, unit prices) and a **total estimated cost**.

3.3 Team Leader

- **Manage team membership** — assign employees to the team or remove them from the team.
- **View team dashboard** — see who is absent today within the team.
- **Team performance overview** — view basic performance indicators per team member (attendance history, review late/absence records, request history, KPIs if exist).
- **Approve/reject team requests** — approve or reject team members' requests (leave/permission).
- **Survey step (review/forward)** — review the survey, add comments and forward the survey to the **Unit Leader**.
- **Escalate to HR** — escalate exceptional cases (balance exceptions or policy overrides) to HR for review/override for him or any member in his team.

3.4 Unit Leader

- **Manage teams** — create or remove teams within the unit.
- **Assign employees to teams** — add or move employees between teams within the unit.
- **Assign team leader to each team.**
- **View unit dashboard** — monitor requests and attendance across the unit.
- **Approve/reject team requests** — approve or reject unit members' requests (leave/permission).
- **Survey step (review/forward)** — review the survey, add comments if needed, and forward it to the **Department Manager**.
- **Escalate to HR** — escalate exceptional cases (balance exceptions or policy overrides) to HR for review/override for him or any member in his unit.

3.5 Department Manager

- **Manage units** — create or remove units within the department.
- **Assign employees to specific team within the unit** — add or move employees between teams within the unit in the department.
- **Assign unit leader to each unit within the department.**
- **Assign team leader to each team within units.**
- **View unit dashboard** — monitor requests and attendance across the unit.
- **Approve/reject team requests** — approve or reject unit members' requests (leave/permission).
- **Survey step (review/forward)** — forward survey to VP or CEO if no VP.
- **Escalate to HR** — escalate exceptional cases (balance exceptions or policy overrides) to HR for review/override for him or any member in his unit.

3.6 Vice President (Optional):

- **Manage departments** — create or remove departments related to this VP.
- **Assign employees to specific team within the unit** — add or move employees between teams within the unit in the department.
- **Assign department manager to each department.**
- **Assign unit leader to each unit within the department.**
- **Assign team leader to each team within units within the department.**
- **View unit dashboard** — monitor requests and attendance across the department.
- **Approve/reject team requests** — approve or reject unit members' requests (leave/permission).
- **Survey step (review/forward)** — forward survey to CEO.
- **Escalate to HR** — escalate exceptional cases (balance exceptions or policy overrides) to HR for review/override for him or any member in his unit.

3.7 CEO:

- **Manage company hierarchy — create or remove hierarchy levels within the company (departments / units / teams).**
- **Manage departments — add new departments or remove one within the company.**
- **Assign Vice Presidents — assign one or multiple Vice Presidents to oversee departments.**
- **Assign Department Managers — assign Department Managers directly if there is no VP.**
- **Assign unit leader to each unit within the department.**
- **Assign team leader to each team within units within the department.**
- **View company dashboard — monitor requests, attendance, and employee activity across the entire company.**
- **Approve/reject high-level requests — approve or reject requests escalated from VP or Department Managers (leave exceptions / purchase orders / loans / assignments).**
- **Survey step (final approval) — review and approve or reject surveys as the final approval step.**
- **Escalate to HR — escalate exceptional cases (policy overrides, disputes, or special approvals) to HR for review or policy enforcement across the company.**

3.8 HR:

- **Manage employees (create/update/activate/deactivate)** — onboard a new employee and assign a team leader in a specific team.
- **Manage employee profile updates** — approve and apply phone/address change.
- **Manage attendance corrections** — fix missing clock-out due to device issues.
- **Manage leave balance & yearly vacation days (per company)** — set annual entitlement to 21 days.
- **Review/override leave requests** — approve leave even if balance is low (policy-based).

- **Create warnings & deductions** — create a late warning and link a deduction to payroll.
- **Manage salary packages** — add transport allowance effective next month.
- **Publish announcements (global or targeted)** — send a notice to one department only.
- **Handle complaints** — assign a complaint to HR staff and mark it resolved.
- **Manage medical class (A/B/C) & insurance policy mapping** — move employees to medical class B.
- **Manage company hierarchy — create or remove hierarchy levels within the company (departments / units / teams).**
- **Survey monitoring (ensure routing completes)** — follow survey steps until the CEO completes it.

3.9 Assets Admin:

- **Maintain asset inventory** — add new assets and update availability status.
- **Handle replace/damage cases** — inspect damaged assets and create replacement action.
- **Process asset returns** — receive returned device and confirm condition.
- **Fulfill asset issue requests** — assign an available laptop (asset_id) after approval.
- Print Serial number of any asset
- Can categorize the assets with count

Mobile app:-

Flutter , Bloc , hive_ce , MVVM , freezed , flutter_animate , skeletonizer , go_router

View_model -> repository -> network service

```
lib/
├── core/
│   ├── network/        # Network Service (Dio/Http client)
│   ├── navigation/      #  App router , Navigation bloc , navigation state
│   ├── constants/       # App constants, API endpoints
│   └── widgets/        # Truly global widgets (e.g., CustomButton)
│   └── utils/           # shared functions
│   └── provider/        # global states like authentication status, theme provider , toast provider
├── data/
│   ├── models/         # Your Entities/DTOs from API
```

```
│   └── repositories/     # (Optional) Logic to choose between API/Local Cache
├── features/           # Replacing "layout" with "features" is standard
│   └── auth/
│       ├── view/        # Auth_view
                           Login
│       ├── view_model/   # Auth_view_model
│       ├── view_loading/ # Auth_Skeleton_loading widgets
│       └── widgets/      # Widgets specific ONLY to Auth
└── main.dart
```

4) Sequence Diagrams:

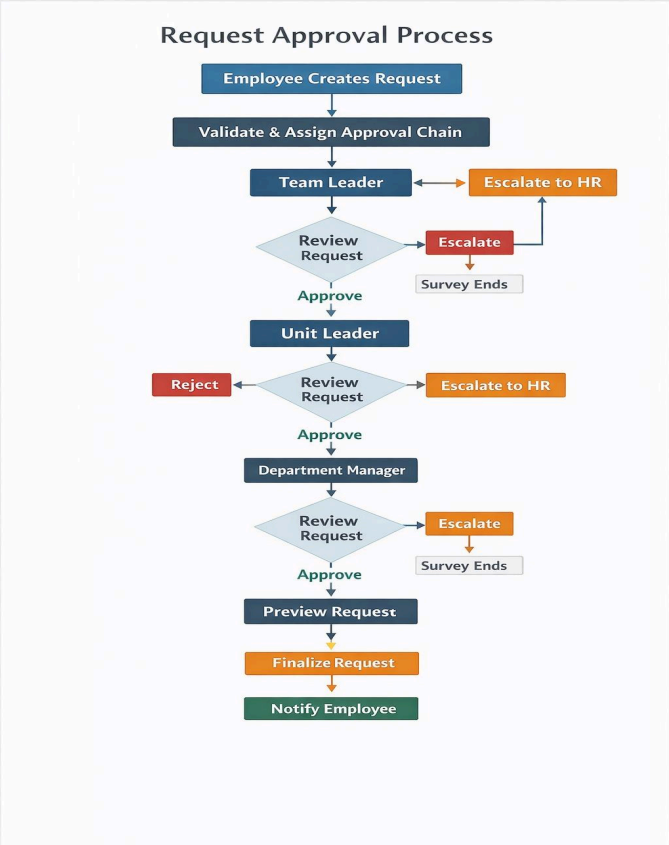
Sequence Diagram 1: Employee Leave Request and Approval

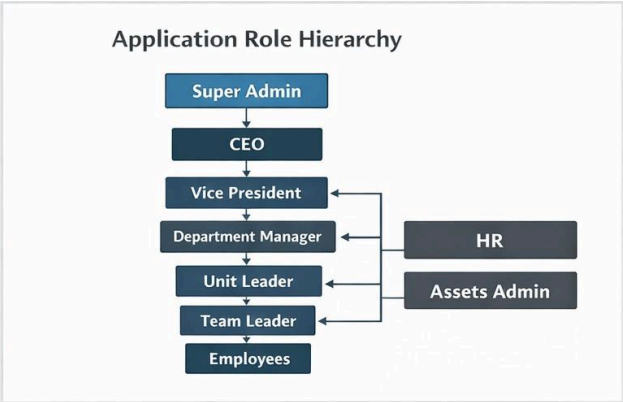
Step	Participant	Action	Recipient	Details
1	Employee	**Create Request**	HRMS S…	Specifies `leave_type`, `start_datetime`, `end_datetime`, `duration`.
2	HRMS Sy…	**Validate & Check Balance**	System	Checks `leave_type` (e.g., annual) against remaining `leave balance`.
3	HRMS Sy…	**Initiate Approval Chain**	Team Le…	Creates `request_id`, sets status to 'in-progress', sends notification.
4	Team Lea…	**Review Request**	HRMS S…	Views request details and team member's history.
5	Team Lea…	**Approve / Reject / Escalate**	HRMS S…	*If Approved*, forwards to Unit Leader. *If Exception*, escalates to HR.
6	HRMS Sy…	**Route to Next Approver**	Unit Lea…	If approved by Team Leader, sends notification to Unit Leader.
7	Unit Leader	**Approve / Reject**	HRMS S…	Repeats Step 4/5 logic, forwards to Department Manager.

8	CEO / Ap…	**Approve High-Level/Exceptions**	HRMS S…	CEO/VP/Dept Manager approves based on the escalation/request type.
9	HRMS Sy…	**Finalize Request**	System	Updates `employment_status`, deducts from `leave balance`.
10	HRMS Sy…	**Notify**	Employee	Sends final notification (Approved/Rejected) and updates `View leave balance & history`.

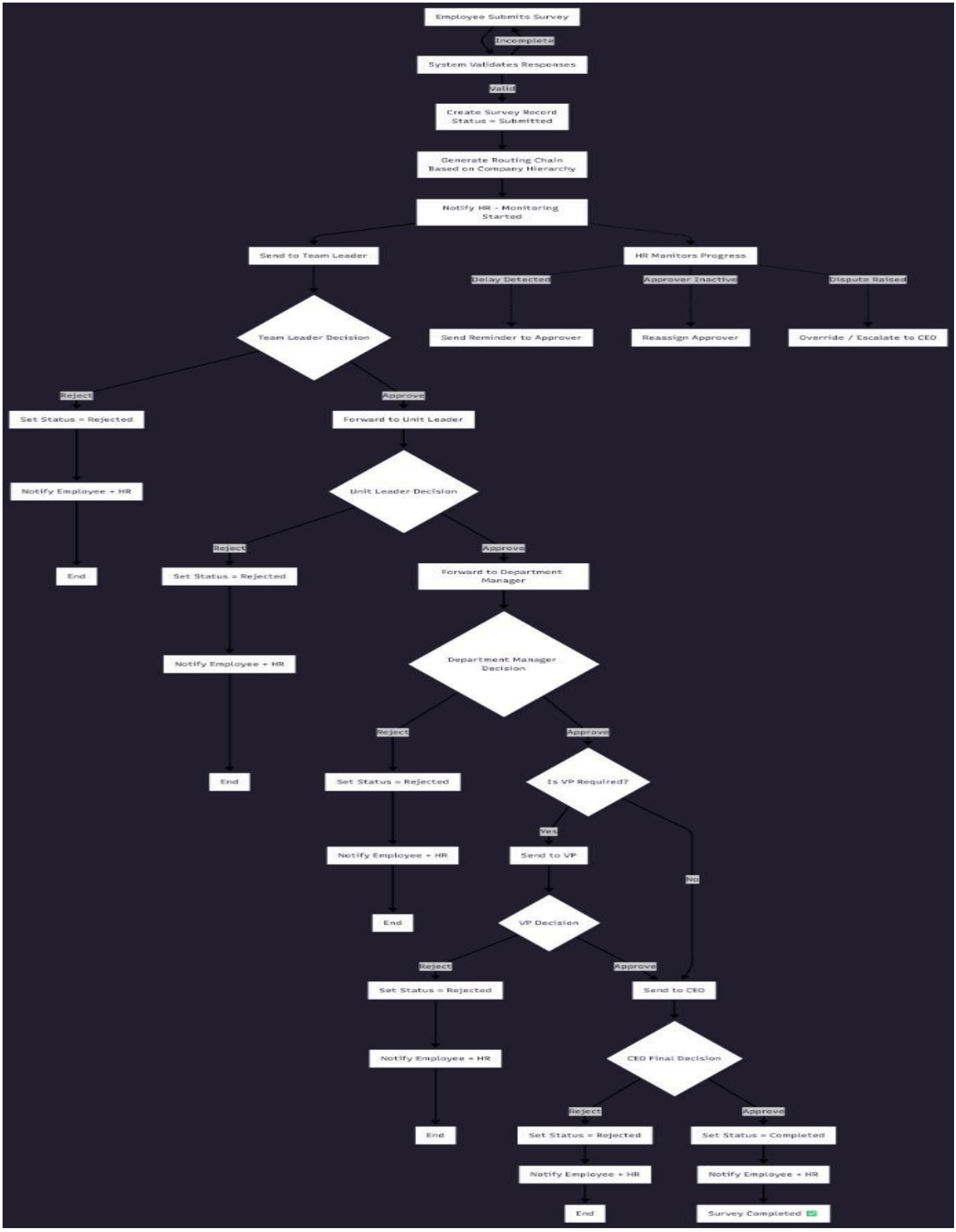
Sequence Diagram 2: Employee Survey Submission and Routing

Step	Participant	Action	Recipient	Details
1	Employee	**Submit Su…**	HRMS…	Submits responses to questions.
2	HRMS Sy…	**Start Rout…**	Team …	Sets survey status to 'submitted', sends notification to Team Leader.
3	Team Lea…	**Review & …**	HRMS…	Reviews, adds optional `comment`, and forwards to Unit Leader.
4	HRMS Sy…	**Route to …**	Unit L…	Sends notification.
5	Unit Leader	**Review & …**	HRMS…	Reviews, adds optional `comment`, and forwards to Department Manager.
6	Dept Man…	**Review & …**	HRMS…	Forwards to VP or CEO if no VP.
7	CEO	**Final Appr…**	HRMS…	Reviews the entire chain and makes the final decision.
8	HRMS Sy…	**Complete …**	System	Sets final status ('completed' or 'rejected').
9	HRMS Sy…	**Notify**	Emplo…	Sends notification of final status/comments.
10	HR	**Monitor**	HRMS…	Follows up to ensure `Survey monitoring (ensure routing completes)`.


Request Approval Process
Employee Creates Request
Validate & Assign Approval Chain
Team Leader
Escalate to HR
Review Request
Escalate
Survey Ends
Approve
Unit Leader
Reject
Review Request
Escalate to HR
Approve
Department Manager
Review Request
Escalate
Survey Ends
Approve
Preview Request
Finalize Request
Notify Employee


Application Role Hierarchy
Super Admin
CEO
Vice President
Department Manager
Unit Leader
Team Leader
Employees
HR
Assets Admin

Survey Approval Process
Employee Submits Survey
Submission Validated
Reject
Notify Employee
Survey Ends
Review by Team Leader
Approve & Forward
Review by Unit Leader
Reject | Notify Employee
Survey Ends
Dept Manager / VP
Approve & Forward
CEO
Final Approval
Survey Completed
Notify Employee


Employee Submits Survey
Incomplete
System Validates Responses
Valid
Create Survey Record Status = Submitted
Generate Routing Chain Based on Company Hierarchy
Notify HR - Monitoring Started
Send to Team Leader
HR Monitors Progress
Delay Detected
Approver Inactive
Dispute Raised
Send Reminder to Approver
Reassign Approver
Override / Escalate to CEO
Team Leader Decision
Reject
Approve
Set Status = Rejected
Forward to Unit Leader
Notify Employee + HR
End
Unit Leader Decision
Reject
Approve
Set Status = Rejected
Forward to Department Manager
Notify Employee + HR
End
Department Manager Decision
Reject
Approve
Set Status = Rejected
Is VP Required?
Notify Employee + HR
Yes
Send to VP
No
End
VP Decision
Reject
Approve
Set Status = Rejected
Send to CEO
Notify Employee + HR
CEO Final Decision
End
Reject
Approve
Set Status = Rejected
Set Status = Completed
Notify Employee + HR
Notify Employee + HR
End
Survey Completed